Iterative Decomposition

Decomposing a bundle into its constituents

Decompose a bundle by repeatedly feeding it into an item memory. Each extracted item is removed from the recurrent network via inhibitory feedback.

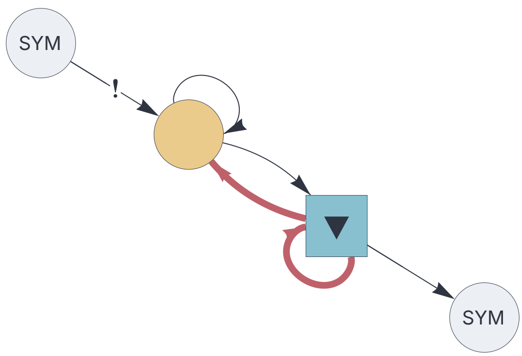
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",

  "hyperparameters": {"default": [2000, 10]},

  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},

    {"component": "delay", "send": [21], "receive": [[{"slot": 11, "tags": ["priority"]}, 21, {"slot": 31,
"tags": ["inhibit"]}]]},

    {"component": "auto", "send": [31], "receive": [[21, {"slot": 31, "tags": ["inhibit"]}]], "decimate":
0.8},

    {"component": "output", "plugin": "codec", "receive": [31]}
  ]
}
```

```
circuit = Circuit[json];
f = circuit["function"];
circuit["schematics"][ImageSize→ 280]
```



Train

```
items = CharacterRange["A", "Z"];
f /@ items;
f[Null]; (* flush delay *)
```

Analyze

```
f[{"A","B","C","D","E","F","G"}]; (* feed bundle of 7 items *)
```

```
Table[ f[Null], 16] (* trigger cycles with empty inputs *)
```

```
{A, D, G, Null, C, E, B, Null, Null, Null, Null, Null, Null, Null, Null, Null}
```